

SMARTASSIST: An Intelligent Email Triage and Response Automation Platform

Bhavani M,
Assistant Professor,

Department of Computer Science and Engineering,
Rajalakshmi Engineering College,
Thandalam, Chennai, India.
bhavani.m@rajalakshmi.edu.in

Logapriya S G,
Student,

Department of Computer Science and Engineering,
Rajalakshmi Engineering College,
Thandalam, Chennai, India.
230701164@rajalakshmi.edu.in

Ragul A,
Student,

Department of Computer Science and Engineering,
Rajalakshmi Engineering College,
Thandalam, Chennai, India.
230701252@rajalakshmi.edu.in

Divya K,
Student,

Department of Computer Science and Engineering,
Rajalakshmi Engineering College,
Thandalam, Chennai, India.
230701082@rajalakshmi.edu.in

Abstract—Customer support staff in enterprises receive an abundance of emails, which requires manual handling. Emails have to be inspected, categorized, prioritized, assigned to agents to send suitable response. These aspects may cause delayed customer service responses, skewed distribution of workload across support agents, and inconsistency in handling customer requests. To tackle the described problems, this paper suggests SMARTASSIST, a solution that implements intelligent automation of support tickets triaging and response generation in the customer support and IT service management domain. Proposed solution comprises email monitoring, ticket creation, intent prediction, ticket priority and department prediction, intelligent ticket routing, retrieval-augmented generation, service level agreement monitoring, and human-in-the-loop approval. Intent, ticket priority and destination departments are predicted for incoming support emails using DistilBERT-based models. Deterministic routing service checks predictions accuracy and assigns tickets to eligible agents. For generating meaningful response, retrieval-augmented generation is used for retrieving standard operating procedures and SLA rules specific to the particular workspace. Workspace-relevant retrieved content is used as context in Llama-based language model for generating response. The generated response is shown to the authorized support agent, who can review, edit, approve or escalate the message before forwarding it back to the customer using the connected email provider. Proposed solution also provides such features as multi-tenancy, workspace isolation, role-based access control, real-time notifications, audit logging, analytics and SLA monitoring. SMARTASSIST combines machine learning, knowledge retrieval, and human oversight to improve support efficiency and consistency while reducing manual effort.

Keywords—Email Triage, Customer Support Automation, IT Service Management, Natural Language Processing, DistilBERT, Retrieval-Augmented Generation, Large Language Models, Intelligent Routing, Load Balancing, SLA Management, Multi-Tenant SaaS.

I. INTRODUCTION

Email has become the primary way of communication between an organization and its customers for resolving any customer queries, problems, technical issues, requests, and performing IT support activities. In the customer support and ITSM environment, organizations receive lots of emails at their support addresses, and each email must be processed in order to comprehend a customer issue, prioritize it, determine the right department and support agent to solve the problem and reply promptly. Processing of all such actions manually can take a lot of time and result in inconsistent prioritization and allocation of support work among support agents. The majority of email-based support solutions requires agent's intervention in reading and categorizing all incoming messages. Such process implies comprehension of message intention, priority setting, standard operation procedure search for resolving the problem and creating the response. As the amount of support tickets increases, manual handling becomes more and more inefficient. Moreover, random allocation of tickets among support agents cannot guarantee a proper load distribution among agents.

New trends in NLP, transformers, and LLMs allow for automation of email support workflow. The use of transformer architectures like DistilBERT allows for classification of emails in terms of their intentions, urgency, and department. At the same time, the enterprise level email support workflow not only entails the classification process, but it should also involve the validation of predictions in accordance with the workplace settings and ticket routing depending on the agent's availability and workload. LLMs can produce high-quality replies; however, they may not have access to organization-specific SOPs, SLAs, and procedures.

In order to overcome the above-listed obstacles, this research proposes a system named SMARTASSIST that is an Intelligent Email Triage and Response Automation Platform. This system observes support email accounts that are connected to the system and normalizes all incoming messages. These normalized messages are transformed to tickets and stored in a PostgreSQL database. DistilBERT models identify the intent of a message, ticket priority, and ticket department. The deterministic routing service confirms the results of prediction and routes the ticket to the available agents for processing based on workload distribution. RAG pipeline obtains relevant SOPs and SLAs for a particular workspace and passes them to the Llama-based language model together with the client's message, context of conversation and predictions to generate a reply to a customer. The reply is then reviewed, edited, approved, or escalated by the support agent.

II. LITERATURE REVIEW

Artificial Intelligence has increasingly been applied to customer support, particularly for ticket classification, priority prediction, and response generation. Bonechi et al. [1] compared BERT-based approaches with traditional machine learning for customer support ticket classification, while Truss and Böhm [2] explored AutoML for ticket category and escalation prediction. Ahmed et al. [3] compared traditional machine learning, deep learning, and transformer-based models for IT incident severity prediction. These studies demonstrate the effectiveness of contextual language models in understanding support requests, although their primary focus remains on classification and prediction.

Recent research has extended ticket automation through knowledge retrieval and generative AI. Xu et al. [4] proposed a Retrieval-Augmented Generation approach combined with Knowledge Graphs for customer service question answering. Nowak-Brzezińska and Drabek [5] introduced a hybrid knowledge-based and machine learning approach for ticket classification, while Jain et al. [6] demonstrated the use of transformer models, semantic embeddings, and RAG for intelligent ticket management. These approaches highlight the benefits of combining AI techniques, although some focus on specific support environments or require complex knowledge representations. Transparency, data quality, and advanced classification have also received attention. Licina et al. [7] proposed an explainable framework for helpdesk ticket classification, emphasizing transparent predictions and human involvement. Fuchs et al. [8] examined human and automated approaches for creating labeled datasets, highlighting the importance of high-quality training data.

Carroll et al. [9] evaluated large language models for automated ticket triage and response generation, demonstrating the benefits of combining LLMs with RAG-based knowledge retrieval. Zangari et al. [10] investigated multi-level ticket classification using transformer-based approaches, while Razma and Jurkus [11] compared traditional machine learning with multilingual transformer models for IT support requests. These studies indicate that support automation requires both accurate classification and

appropriate handling of complex, organization-specific requests. The application of AI to email processing has also evolved from traditional statistical approaches to modern deep learning models. Dredze et al. [12] presented an early study on intelligent email processing using LSA and LDA, while Victoire and Nithyalakshmi [13] proposed an AI-driven system for centralized email triage and SLA management.

Although these studies demonstrate the potential of automating email processing and ticket handling, many existing approaches focus on individual stages rather than a complete end-to-end support workflow. SMARTASSIST addresses this gap by integrating email ingestion, DistilBERT-based intent and priority classification, department routing, workload-balanced agent assignment, and RAG-based response generation. The system further uses pgvector for semantic knowledge retrieval and a LLaMA-based model for contextual response drafts, while retaining human supervision for final approval.

III. PROPOSED SYSTEM

The proposed model follows an end-to-end intelligent workflow for support services, from customer email collection to delivery. It combines natural language processing along with deterministic business rules to assist support operations without allowing AI to make critical decisions independently. The workflow includes email ingestion and preprocessing, ticket creation, AI classification, entity extraction, routing, agent assignment, RAG retrieval, response generation, human approval, and email delivery. The backend service layer connects these processes, while PostgreSQL stores emails, tickets, predictions, assignments, knowledge-base data, and audit records. The complete processing workflow is illustrated in fig.1.



Fig.1. End-to-End Workflow

A. AI-based Email Intent Classification and Priority Prediction

The email is passed through the classification pipeline in which DistilBERT processes the text content of the support request. DistilBERT is a lighter version of the BERT language model and gives context-based interpretation of

the language at reduced computational costs compared to the original BERT model. Classification is done on two grounds – intent of the request and priority of the request. The model is trained to classify 25 intent which includes password reset, login issues, network, hardware, software/applications, security, and IT general. Priority is classified as LOW, MEDIUM, HIGH, or CRITICAL. The following fig.2. illustrates the email classification workflow.

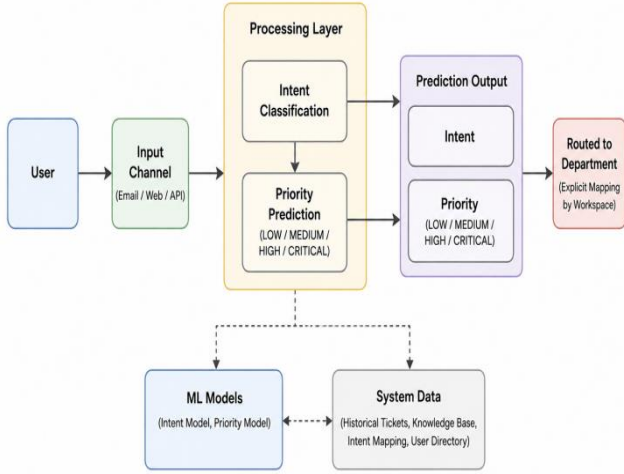


Fig.2.Email Classification Workflow

In addition to classification, the system uses machine learning-based entity extraction, structured parsing, and deterministic rules to identify entities in the emails. These entities may include the name and email address of the customer, ticket ID or reference, product or application, system name, error message, date and time, money value, and description of the issue which are included in the emails sent by the customer. The predicted labels, confidence scores, and identified entities is stored in the database for analysis and ticket processing purposes. However, the prediction made by DistilBERT is not used to assign tickets to any agent. The output of the model is provided to the deterministic routing service.

B. Intelligent Routing

After obtaining the intent, priority, and confidence score, the routing service assigns the ticket using the workspace's department–intent mappings. The intent that has been predicted is mapped against the department to determine the correct department to route the ticket, after which the availability of support personnel in the correct department for the given priority ticket is determined. In the case where the confidence score is lower than the required threshold, then the ticket is sent to the general support or review queue. This separation allows AI predictions to work alongside organization-defined routing rules and workspace-specific support structures.

C. Load-Balanced Agent Assignment

Once the department is identified, the routing service assigns the ticket to an eligible support agent using a workload-aware policy. It calculates each agent's workload

based on the number and priority of their active tickets, giving greater weight to high- and critical-priority tickets:

$$W_i = O_i + \alpha H_i + \beta C_i$$

where W_i represents the workload of agent i , while O_i , H_i , and C_i denote the agent's open, high-priority, and critical-priority tickets, respectively. The agent with the minimum workload is selected:

$$\text{AgentSelected} = \text{Minimum}(W_i)$$

If multiple agents have the same workload, the senior agent who joined the workspace earlier is selected as the tie-breaker. The assignment is performed within a database transaction to ensure consistency when processing multiple tickets simultaneously.

D. RAG Based SOP And Knowledge retrieval

The proposed system uses a workspace-specific knowledge base containing SOPs and approved response templates. The documents that are uploaded undergo the following stages: text extraction, segmentation into parts and conversion into vectors with the help of Sentence Transformer model. The document embedding process is illustrated in Fig. 3. These embeddings are stored in PostgreSQL with the pgvector extension.

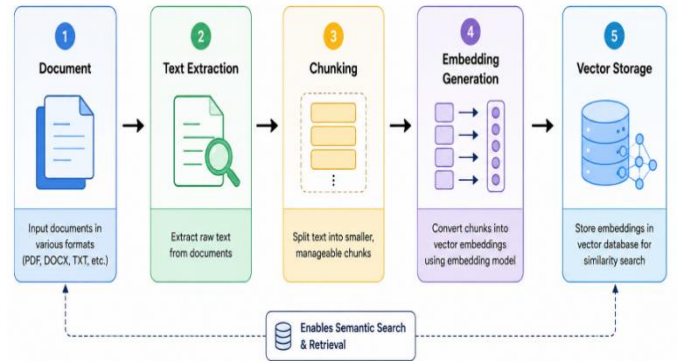


Fig.3.Document Embedding Pipeline

The process of dealing with a support ticket involves the creation of the embedding of the ticket and matching it with the document vectors, in order to find the most related chunks of knowledge. The retrieval is limited to the workspace of the user who is logged into the system. Department and intent are considered when finding the most related chunks of knowledge. The SOPs and templates are fed into the answer generation module.

E. SLA Policy Retrieval and Management

The suggested system allows setting the policy related to the Service Level Agreement (SLA) for LOW, MEDIUM, HIGH, and CRITICAL tickets for each workspace. Once the ticket is created, the appropriate policy is determined, and the deadlines for response and resolution based on the ticket priority are calculated. SLA information is attached to the ticket and constantly monitored through the background process. The SLA is evaluated according to the current status of the ticket and time that has passed since its creation. Thus, the SLA could be considered as

ON_TRACK, AT_RISK, BREACHED, or MET. Notifications are sent to the assigned agent, and to the manager if needed.

F. Llama-Based Response Generation

The process of classification, information extraction, routing, and knowledge retrieval is followed by the generation of an agent-ready response by a Large Language Model running on Llama. In this process, the model uses the original email and dialogue transcript, intent, department, information extracted, SOPs, SLA information, and approved templates to generate a professional answer solely based on the information presented without making up policies, dates, actions, or commitments. If there is not enough information on organizational issues, then it creates a conservative draft for review and approval. Fine-tuned and pre-trained Llama model is preferred to avoid any security issues as the response generation module will take company-specific data like SLA policies, SOPs as input to generate more relevant and specific response.



Fig.4. Response Generation

G. Human-in-the-Loop Approval

The proposed framework provides for complete validation of all AI-generated responses before delivery to the customers. The agents have access to editing, refinement, and finalizing the draft right in the ticket. The tickets requiring further details or approval can be escalated to the managers for further analysis. At the same time, there will be a running automated process of security procedures in the background right before the ticket is sent. The procedure includes checking for the correct user authorization, permissions for the workspace and ticket, recipient identity, Gmail credentials, and the ticket itself.

By integrating mandatory human editorial review with rigorous automated system checks, this workflow guarantees that unapproved, inaccurate, or hallucinated information never reaches the client. As a result, such a balance allows maintaining the speed of work along with the necessary transparency and privacy.

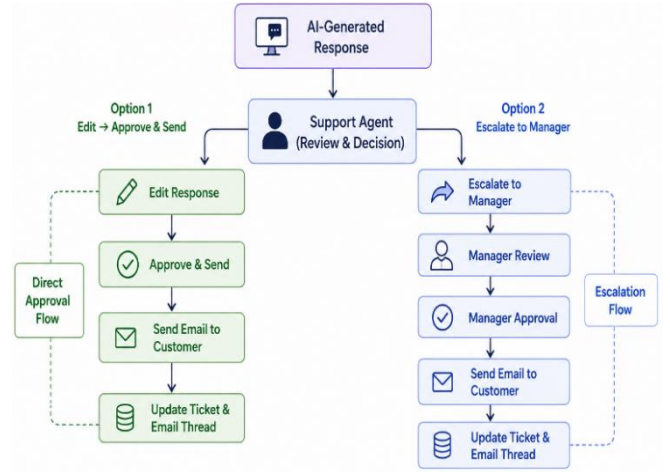


Fig.5. Review and Response by Support Agent

H. Multi-Tenant Security and Authorization

The designed architecture consists of a multi-tenant architecture where each workspace serves as an isolated environment for a particular business. Users can make a new workspace on their own as an owner of the workspace or become part of any workspace with an authorized invitation. Each user is part of one workspace, but email addresses, emails, tickets, departments, SOPs, and any AI-related information are linked with the respective workspace. Access is limited depending on the membership of the workspace and roles assigned to a user, such as Owner, Manager, and Support Agent.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The proposed system is a multi-tenant modular platform integrating email processing, AI-based ticket analysis, knowledge retrieval, SLA management, and human-supervised response handling. Incoming emails are converted into structured tickets, classified, routed, and assigned, while RAG retrieves relevant organizational knowledge and SLA policies monitor deadlines. Generated responses are reviewed and approved by support agents before delivery. This section presents the system architecture, implementation technologies, database design, and workspace-specific access control.

A. System Architecture

The proposed system uses a modular architecture combining email processing, natural language understanding using DistilBERT model, knowledge retrieval, ticket management, and human approval. Incoming support emails are converted into uniquely identified tickets, which undergo intent classification, entity extraction, priority prediction, department routing, and agent assignment. The RAG module retrieves relevant SOPs and response templates, while the applicable SLA policy determines response and resolution deadlines.

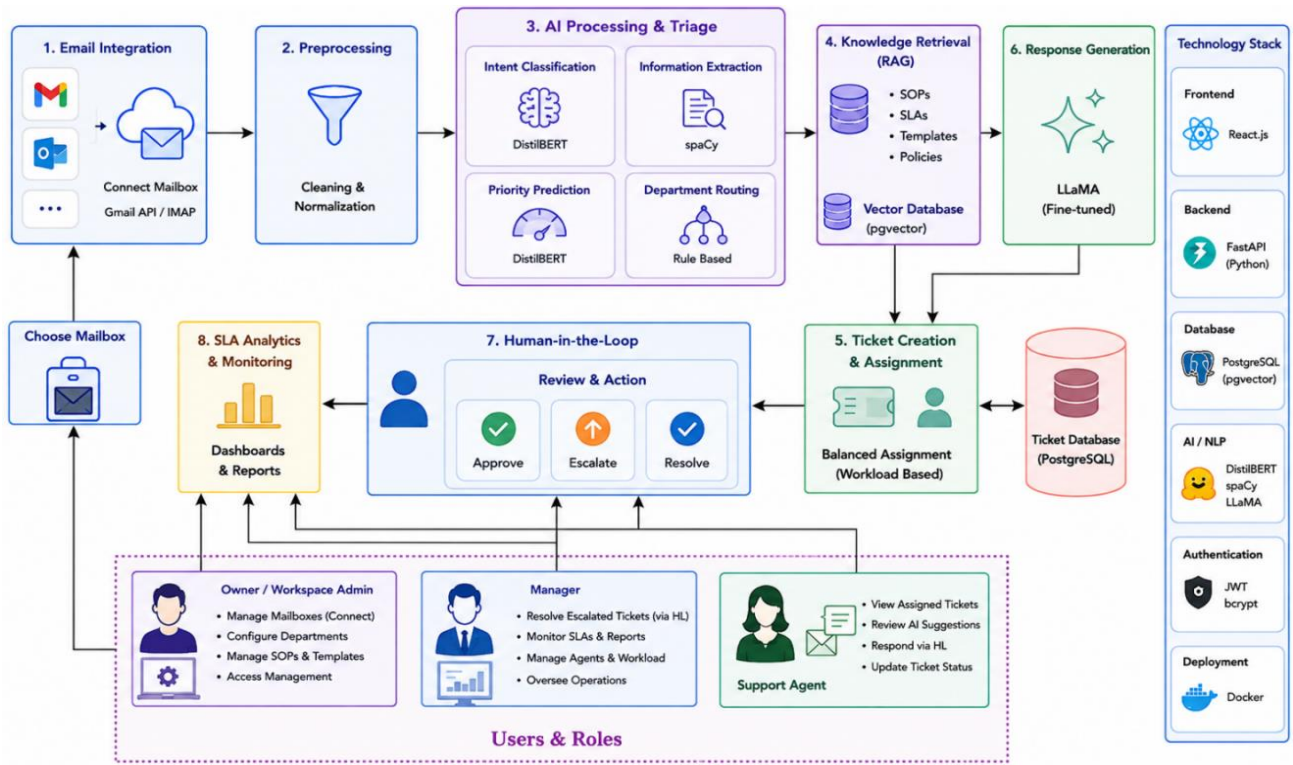


Fig.6.Architecture Diagram.

The response generation module receives the ticket context and retrieved knowledge which are then reviewed by the support agent who can edit, approve, or escalate the response to the manager if needed before delivery. AI predictions remain separate from organizational business rules and are subject to human oversight.

B. Technologies Used

The described table in fig.7 illustrates the technologies, model and framework used to implement the proposed system. It uses React and FastAPI for the user interface and backend API development. PostgreSQL with pgvector was selected to manage ticket data in a single database. A fine-tuned multitask DistilBERT model was chosen for its efficient contextual understanding and lower computational cost to classify intent and predict priority. RAG, together with sentence transformers is used for creating embeddings and semantic retrieval of knowledge base. Llama generates contextual response drafts. Gmail API provide email integration.

Component	Technology
Frontend	React
Backend	FastAPI
Database	PostgreSQL
Vector Storage	pgvector
Intent Classification	DistilBERT
Priority Prediction	DistilBERT
Entity Extraction	spaCy
Text Embeddings	Sentence Transformer
Knowledge Retrieval	RAG
Response Generation	Llama
Email Integration	Gmail API / Microsoft Graph API
Authentication	JWT

Fig.7.Technologies Used

C. Database Design

The SmartAssist database uses PostgreSQL to store users, workspaces, email accounts, tickets, routing information, and roles. Primary and foreign keys maintain data consistency and support tracking throughout the ticket lifecycle. The Users, Workspaces, and Workspace Memberships tables manage workspace membership and role-based access for Owners, Managers, and Support Agents.

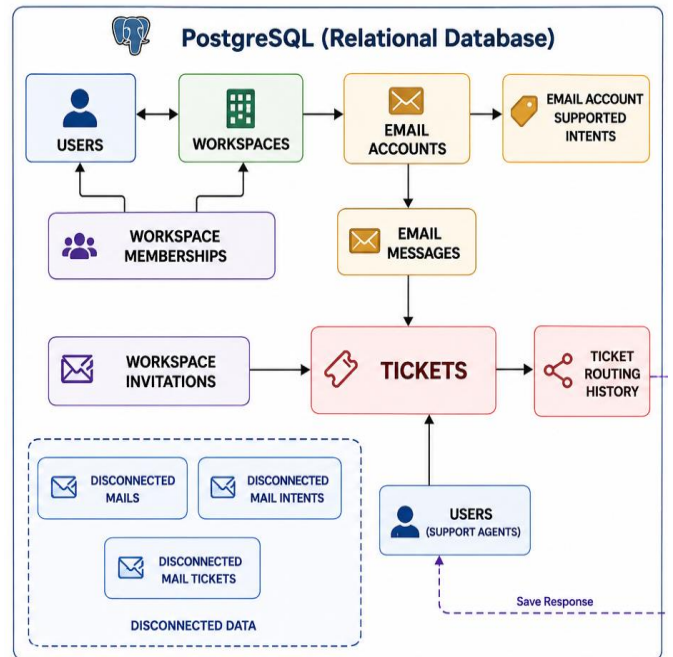


Fig.8.Database Design

Email Accounts stores mailbox details, while Email Messages and Tickets manage incoming emails and ticket information. Ticket Routing History records routing changes, and assigned_agent_id supports workload balancing. Workspace Invitation manages user invitations, while the disconnected-mail tables preserve related data when a mailbox is disconnected.

D. RAG Knowledge Base And PGvector

SmartAssist uses a Retrieval-Augmented Generation (RAG) model to generate context-aware responses using SOPs, SLAs, policies, FAQs, support information, and resolved tickets. The documents are broken down into smaller chunks and then represented as vector embeddings that are indexed and searched based on semantic similarities using the pgvector extension in PostgreSQL. In generating a response to the ticket, the related knowledge is extracted as context and fed to the large language model to generate an organization-specific response draft.

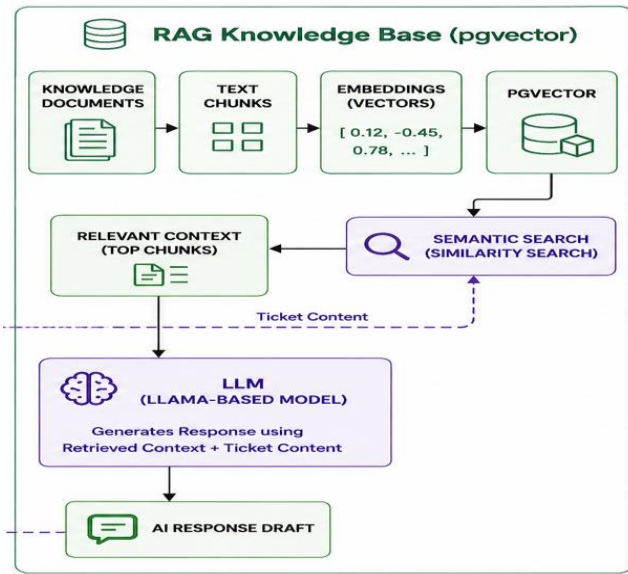


Fig.9.Pgvector Design.

V. EXPERIMENTAL SETUP

A. Dataset Collection

In total, the SmartAssist dataset has 5,000 IT customer support emails with content, body, intent, and priority attributes. There are 25 intent classes and 3 priority classes (High, Medium, Low). Content and body were merged to keep both the summary of the request and the context. A publicly available customer-support email dataset was obtained from Kaggle and used for model training. The dataset was preprocessed and adapted to the predefined intent and priority categories of SmartAssist such that it has no missing or duplicated values. Intent and priority labels have been encoded and combined stratification has been applied to keep the distribution when splitting data. Dataset split ratio is 70%, 15%, and 15% for training, validation, and test sets resulting in 3,500, 750, 750 emails for each training, validation and test sets respectively. The text has been tokenized by distilbert-base-uncased tokenizer with max_length equal to 256 needed to understand its meaning and priority.

Intent	Intent	Intent
Account Access	Password Reset	Access Permission
Account Locked	Payment Failed	VPN Issue
Application Error	Refund Request	Network Connectivity
Billing Issue	Security Incident	Email Issues
Complaint	Service Outrage	Slow Performance
Duplicate Payment	Software Bug	Data Export Request
Feature Request	Subscription Issue	Two Factor Authentication
General Support	Invoice Issue	Password Issue
Login Issue	-	-

Table.1.Support Request Intent Categories Used for Classification

B. Dataset Processing

Before classification, email content is cleaned by removing HTML elements, unnecessary whitespace, repeated content, irrelevant formatting, and unused headers or metadata. The subject and message body are retained, normalized, and tokenized using the DistilBERT tokenizer. Sequences are padded or truncated to a maximum length of 256 tokens.

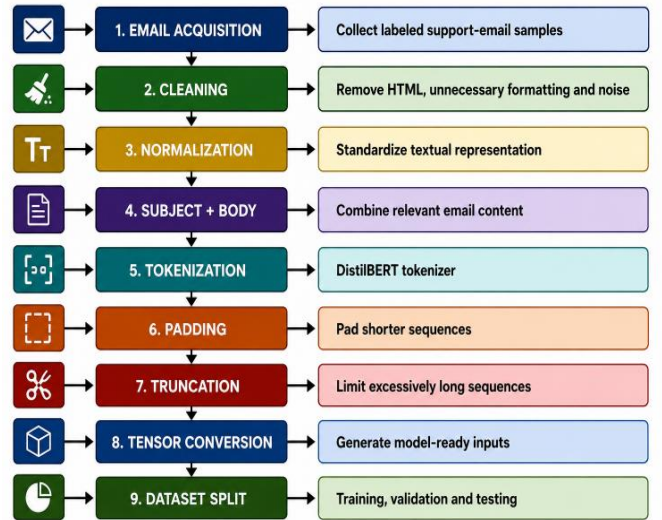


Fig.10.Database Preprocessing Pipeline

C. Model Training

A DistilBERT-based multi-tasking model was developed for predicting the intention and priority of emails from the IT support. The distilbert-base-uncased model works as the text encoder and is followed by two classification heads, one for the intents having 25 classes and another for the priority with 3 classes. A dropout of 0.2 was used in order to prevent overfitting. Intent and priority cross-entropy losses were added together to form the loss during training. The model was trained for 5 epochs using a batch size of 16, a learning rate of 2×10^{-5} and weight decay of 0.01. FP16 training was carried out on a Tesla T4 NVIDIA GPU which are illustrated in table 2.

Dataset Split	Percentage
Base Model	DistilBERT (distilbert-base-uncased)
Architecture	Multi-task DistilBERT
Intent classes	25
Priority classes	3
Training Samples	3500
Validation Samples	750
Test Samples	750
Epochs	5
Batch Size	16
Learning Rate	2×10^{-5}
Weight Decay	0.01
Dropout	0.2
Maximun Sequence Length	256 tokens
Optimizer/training Framework	Hugging Face Transformers / PyTorch
Hardware	NVIDIA Tesla T4 GPU

Table.2.Distribution of Dataset Across Training, Validation, and Testing Sets

VI. RESULTS AND DISCUSSION

SmartAssist combines e-mail-to-ticket mapping, automated classification using AI, deterministic routing, workload-based assignment, RAG-based response composition, and human approval in a comprehensive support process. This chapter examines the classification efficiency and training pattern of the suggested approach.

A. User Interface

SmartAssist offers a role-based interface for handling support tickets. The Workspace Owner handles the management of members and associated emails, the Manager handles the activity of the tickets, agent load, and escalations, whereas the Support Agent handles their assigned tickets and escalations. The other features available through the interface include the prediction of intents, priorities, confidences, and routing.

B. Classification Performance evaluation

The multi-tasking DistilBERT model was validated on validation and test data for intents and priorities classification. Loss from training reduced from 3.6905 to 0.3023 while loss from validation reduced from 2.6634 to 0.2016 within 5 epochs. Evaluation of the model included accuracy, precision, recall, F1 Score, and prediction confidence. On the test set of emails, the average confidence was 86.34% for intent and 95.15% for priority, with minimum confidence values of 72.37% and 88.78%, respectively.

Evaluation	Intent	Priority
Validation Accuracy	85%	89%
Validation Precision	84%	81%
Validation Recall	86%	79%
Validation F1	78%	82%
Validation Average Confidence	86.06%	95.07%
Test Accuracy	88%	82%
Test Average Confidence	86.34%	95.15%
Test Minimum Confidence	72.37%	88.78%
Test Predictions $\geq 70\%$	660/750	615/750

Table.3.Classification Model Evaluation Results

The findings reveal that the multi-task learning method succeeded in understanding both the semantic meaning and the urgency of the support tickets based on the dataset used. Additionally, the confidence measures constitute another form of prediction confidence which will assist in the confidence-based review system employed by SmartAssist. Nevertheless, it should be noted that the performance of the DistilBERT model is constrained to the specific data set. Thus, the performance evaluation of the model should include testing against other real-world support emails.

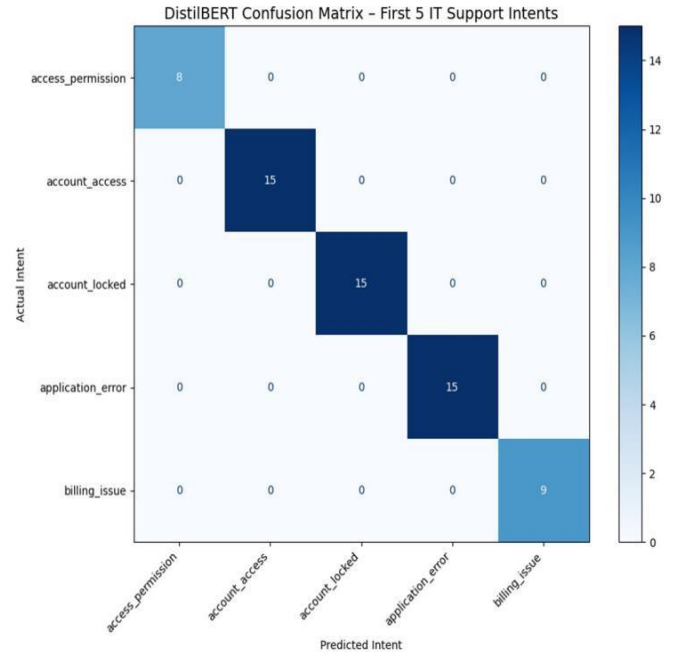


Fig.11.Confusion Matrix for Selected IT Support Intents

C. Training And Validation Analysis

The training and validation losses decreased consistently across the five epochs, indicating effective learning and stable convergence. As shown in Fig. 13, the similar downward trends suggest no significant divergence between

the training and validation sets during training. The reduction in validation loss also reinforces the choice of the last trained model based on validation loss. However, more experiments using other data and training settings would still need to be conducted to test this.

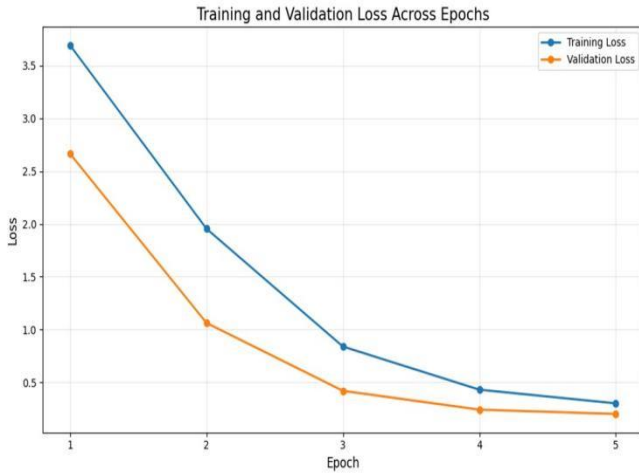


Fig.12. Training and Validation Loss Across Epochs

VII. CONCLUSION

The proposed intelligent email triaging and response automation system provides an integrated approach to automating the key functions of customer support and IT service management. The proposed system involves a transformer-based classification and information extraction from emails, deterministic routing, allocation of agents depending on the workload, workspace isolated retrieval augmented generation, SLA management, and Llama-based response generation process. The segregation of the prediction process based on AI from rule-based routing, along with human intervention before any contact with a customer, allows not only to increase the level of automation but also to provide proper control and security. The workspace isolation combined with role-based access allows providing multi-tenancy, and the continuous email monitoring, instant notifications, and SLA tracking allow managing incoming emails properly. The usage of the retrieval-augmented generation mechanism helps ground the answers to organization-specific SOPs and policies. Thus, the proposed solution is expected to provide improved automation of the triage process, increased consistency of classification and ticket assignments, support-agent workload balancing, and reduced customer response time while keeping the necessary level of control. mechanism, and provide additional features like automated escalation and analytics of the support emails.

VIII. FUTURE WORK

For future work, the goal will be testing SmartAssist with a bigger and more varied dataset of support emails to increase generalizability of the models. The classification model can be optimized through inclusion of more intents and continuous updates of training data. Another potential area for improvement is to enable the system to work with multiple mailboxes and email clients by enabling ingesting

and tracking of emails across all organizational email addresses. Support across multiple languages and domain-specific entity extraction are other possibilities. In addition, the retrieval and re-ranking in the RAG pipeline can be optimized. Some of the features that could be included in the future are SLA prediction, escalation identification, and workload forecasting.

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